

DiRty Data Done Dirt Cheap

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Highway to… Clean Data!

(lecture roadmap)

Part I: What are our tools?

Part II: What's our dataset?

Part III: Where do we start?

Part IV: Cleaning the columns

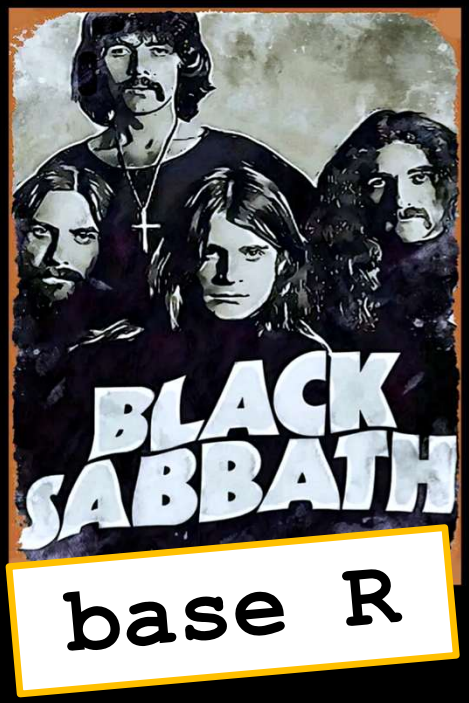


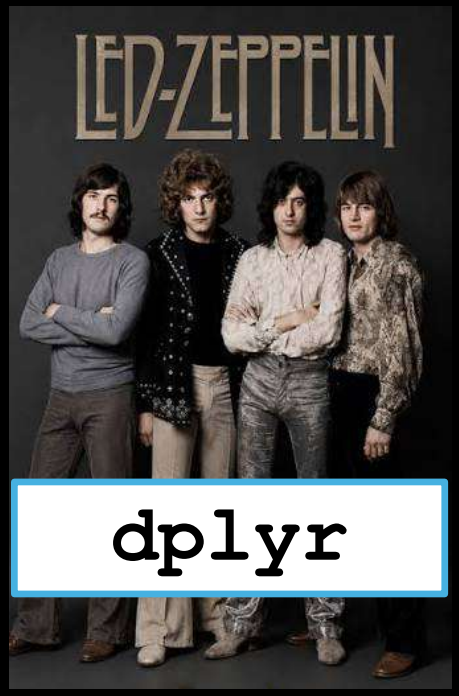
Angus Young, AC/DC

Part I:
What are our tools?



The Unholy Trinity







The main data-cleaning tools for this lecture.

dplyr package

- **When to use it?** Data cleaning, transformation, and manipulation.
- **"Grammar of data manipulation."** Uses intuitive verbs:

filter()	group_by()
pull()	mutate()
summarise()	arrange()
count()	bind_rows()



stringr package

- **When to use it?** Manipulating and exploring character strings.
- Can extract, detect, count, replace, match, and trim character strings.

extract	str_extract()
detect	str_detect()
count length	str_length()
replace	str_replace()
match	str_match()
trim	str_trim()



Part II:
What's our dataset?



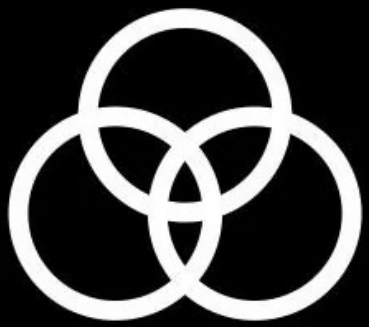
Part II: What's our dataset?

musician_name	instrument_role	band	years_in_band	age	alive	date_of_death	nationality	rock_hall_inductee
ritchie blackmore	guitar	Deep Purple	1968-1975	0	1		ENGLAND	TRUE
JOHN PAUL JONES	Bass/Keyboards	Led Zepplin	1968-1980	80	Y		ENGLAND	TRUE
David Coverdale	Vocal	Whitesnake	1978 - Present	74	1		english	no
James Hetfield	Vocals/Guitar	metallica	1981-present	N/A	yes		united states	TRUE
ronnie james dio	Lead Vocals	Dio	1982/2010	67	N	5/16/2010	America	Y
JIMMY PAGE	Lead Guitar	Led Zeppelin	1968 to 1980	82	Yes	1/1/2026	United Kingdom	yes
Phil Rudd.	Percussion	ac/dc	1975 - Present	72	alive		Australia	Yes
Kirk Hammett	Guitar	Metallica	1983-current	seventy-eight	Yes		U.S.A.	yes
Bruce Dickinson.	Vocal	IRON MAIDEN	1981-present	150			England	Y
Ritchie Blackmore	guitar	Ritchie Blackmore's Rainbow	'75-'84	-5	TRUE		ENGLAND	YES
Bon Scott.	vocals	Ac/Dc	unknown	33	no	19-Feb-80	UK	TRUE
cliff burton	BASS	metallica	1982 to 1986	24	deceased	27/09/1986	united states	yes
Steve Harris	Bass	Iron Maiden	1975 - Present	70	yes		ENGLAND	Yes
Ian Gillan	VOCALS	Black Sabbath	1983/1984	81	1		U.K.	Y
Robert Plant.	VOCALS	Led Zepplin	1968-1980	78	Y		United Kingdom	TRUE
Roger Glover.	Bassist	deep purple		80	yes		England	TRUE
DAVID COVERDALE	singer	Deep Purple	1973-1976	74	alive		U.K.	Yes
Tony Iommi	Guitarist	Black Sabbath	1968-2017	78	Yes		England	yes
Ian Paice	DRUMS	Deep Purple	1968-present	78	alive		english	TRUE
Ronnie James Dio.	Lead Vocals	Black Sabbath	1979-1982	67	N	16-May-10	USA	
Glenn Hughes	Vocals/Bass	Deep Purple	1973/1976	75	alive		U.K.	

View(data_raw)

Part III:

Where do we start?



Where do we start?

Looking at the entire dataset…

STEP 1: Identify Patterns

STEP 2: Find the “Big Stuff”

STEP 1: Identify Patterns

- All data contain patterns… even dirty data
- What's showing up more than once?
- Within and between columns, what's consistently different or alike?



Cinderella

Part III: Where do we start?

musician_name	instrument_role	band	years_in_band	age	alive	date_of_death	nationality	rock_hall_inductee
ritchie blackmore	guitar	Deep Purple	1968-1975	0	1	*NA*	ENGLAND	TRUE
JOHN PAUL JONES	Bass/Keyboards	Led Zepplin	1968-1980	80	Y		ENGLAND	TRUE
David Coverdale	Vocal	Whitesnake	1978 - Present	74	1		english	no
James Hetfield	Vocals/Guitar	metallica	1981-present	N/A	yes		united states	TRUE
ronnie james dio	Lead Vocals	Dio	1982/2010	67	N	5/16/2010	America	Y
JIMMY PAGE	Lead Guitar	Led Zeppelin	1968 to 1980	82	Yes	1/1/2026	United Kingdom	yes
Phil Rudd.	Percussion	ac/dc	1975 - Present	72	alive		Australia	Yes
Kirk Hammett	Guitar	Metallica	1983-current	seventy-eight	Yes		U.S.A.	yes
Bruce Dickinson.	Vocal	IRON MAIDEN	1981-present	150			England	Y
Ritchie Blackmore	guitar	Ritchie Blackmore's Rainbow	'75-'84	-5	TRUE		ENGLAND	YES
Bon Scott.	vocals	Ac/Dc	unknown	33	no	19-Feb-80	UK	TRUE
cliff burton	BASS	metallica	1982 to 1986	24	deceased	27/09/1986	united states	yes
Steve Harris	Bass	Iron Maiden	1975 - Present	70	yes		ENGLAND	Yes
Ian Gillan	VOCALS	Black Sabbath	1983/1984	81	1		U.K.	Y
Robert Plant.	VOCALS	Led Zepplin	1968-1980	78	Y		United Kingdom	TRUE
Roger Glover.	Bassist	deep purple		80	yes		England	TRUE
DAVID COVERDALE	singer	Deep Purple	1973-1976	74	alive		U.K.	Yes
Tony Iommi	Guitarist	Black Sabbath	1968-2017	78	Yes		England	
Ian Paice	DRUMS	Deep Purple	1968-present	78	alive		english	

View(data_raw)

STEP 1: What patterns do you see?

Part III: Where do we start?

STEP 1: What patterns do you see?

Using glimpse() from dplyr

```
> glimpse(data_raw)
Rows: 36
Columns: 9
$ musician_name      <chr> "ritchie blackmore  ", "JOHN PAUL JONES", "David Coverdale", "James  Hetfield", " ronnie james dio",
$ instrument_role    <chr> "guitar", "Bass/Keyboards", "Vocal", "Vocals/Guitar", "Lead Vocals", "Lead Guitar", "Percussion", "G
$ band               <chr> "Deep Purple ", "Led Zepplin", "Whitesnake", "metallica", "Dio", "Led  Zeppelin", "ac/dc", "Metallic
$ years_in_band      <chr> "1968-1975", "1968-1980", "1978 - Present", "1981-present", "1982/2010", "1968 to 1980", "1975 - Pre
$ age                <chr> "0", "80", "74", "N/A", "67", "82", "72", "seventy-eight", "150", "-5", "33", "24", "70", "81", "78"
$ alive              <chr> "1", "Y", "1", "yes", "N", "Yes ", "alive", "Yes", "", "TRUE", "no", "deceased", "yes", "1", "Y", "y
$ date_of_death      <chr> "", "", "", "", "5/16/2010", "1/1/2026", "", "", "", "", "19-Feb-80", "27/09/1986", "", "", "", "",
$ nationality        <chr> "ENGLAND", "ENGLAND", "english", "united states", "America  ", "United Kingdom", "Australia", "U.S.A
$ rock_hall_inductee <chr> "TRUE", "TRUE", "no", " TRUE", "Y", "yes", "Yes", "yes", "Y", "YES", "TRUE", "yes", "Yes", "Y", "TRU
```

STEP 2: Find the "Big Stuff"

- Trashed hotel room analogy
- What will get in the way the most?







Part III: Where do we start?

STEP 2: Find the “Big Stuff”

1. Capitalization inconsistencies
2. Lack of true NA values
3. Similarities between “alive” and “rock_hall_inductee” columns
4. All columns are character vectors
5. Leading and trailing spaces

Part III: Where do we start?

STEP 2: Find the "Big Stuff"

1. Capitalization inconsistencies → Easy way to decrease variation
2. Lack of true NA values → Decreases "blind spots"
3. Similarities between "alive" and "rock_hall_inductee" columns
4. All columns are character vectors
5. Leading and trailing spaces → Easy way to decrease variation

Fixing the "Big Stuff"

Capitalization inconsistencies

- Title case text (*stringr*) → `str_to_title()`
- Lower case text (*stringr*) → `str_to_lower()`

Leading and trailing spaces

- *stringr* → `str_squish()`

Lack of true NA values

- Indexing (*base R*) → `data[data == ""] = NA`
- *dplyr* → `na_if()`

Part III: Where do we start?

Intro Code

```r
#standardizing capitalization
#title case with str_to_title
data_intro$musician_name = str_to_title(data_intro$musician_name)
data_intro$band = str_to_title(data_intro$band)
data_intro$nationality = str_to_title(data_intro$nationality)
#lower case with str_to_lower
data_intro$instrument_role = str_to_lower(data_intro$instrument_role)
data_intro$alive = str_to_lower(data_intro$alive)
data_intro$rock_hall_inductee = str_to_lower(data_intro$rock_hall_inductee)

#removing white space using str_squish function from stringr
#NOTE: str_squish will coerce all variables to be characters.
#That's not a problem for us since all our variables are characters already.
data_intro$musician_name = str_squish(data_intro$musician_name)
data_intro$instrument_role = str_squish(data_intro$instrument_role)
data_intro$band = str_squish(data_intro$band)
data_intro$years_in_band = str_squish(data_intro$years_in_band)
data_intro$age = str_squish(data_intro$age)
data_intro$alive = str_squish(data_intro$alive)
data_intro$date_of_death = str_squish(data_intro$date_of_death)
data_intro$nationality = str_squish(data_intro$nationality)
data_intro$rock_hall_inductee = str_squish(data_intro$rock_hall_inductee)

#turning blank cells into NA
data_intro[data_intro == ""] = NA

#checking work
glimpse(data_intro)
```

Advanced Code

```r
data_adv = data_adv %>%
  mutate(
    #standardizing capitalization
    #title case with str_to_title
    across(c(musician_name,
             band,
             nationality),
           ~ str_to_title(.)),
    #lower case with str_to_lower
    across(c(instrument_role,
             alive,
             rock_hall_inductee),
           ~ str_to_lower(.)),
    #removing white space using str_squish function from stringr
    across(where(is.character), str_squish),
    #turning blank cells into NA
    across(everything(), ~na_if(., "")))

#checking work
glimpse(data_adv)
```

Use the code that makes sense to YOU.
You can't control code you can't understand.

Part IV:

Cleaning the columns



What's our process?

STEP 1: View the data

STEP 2: Verbalize the problems and patterns

STEP 3: Verbalize the fix

STEP 4: Find the code for the fix



AGE <chr>
62
55
Seventy-five
58
78
69
1,000
57
77
-1
59

Part IV: Cleaning the columns

instrument_role

```
> table(data_adv$instrument_role, useNA = "ifany")

          bass bass/keyboards        bassist       drummer         drums        guitar
             4              1              1              1              2             5
     guitarist      keyboards    lead guitar    lead vocals     percussion        singer
             1              1              3              4              2             1
         vocal         vocals    vocals/bass   vocals/guitar
             2              6              1              1
```

STEP 1: View the data

STEP 2: Verbalize the problems and patterns

STEP 3: Verbalize the fix

STEP 4: Find the code for the fix

Part IV: Cleaning the columns

instrument_role

Separating double instruments

- *stringr* → `str_split_fixed()`

"If... Then" criteria

- *dplyr* → `case_when()`

Finding matching character strings

- *stringr* → `str_detect()`



Heart

Part IV: Cleaning the columns

instrument_role

Intro Code

```r
#separating cells with two instruments into two separate columns
#using str_split_fixed from stringr
data_intro[c("instrument_primary", "instrument_secondary")] =
  str_split_fixed(data_intro$instrument_role, "/", n = 2)

#cleaning up instrument labels for using str_detect() from stringr
#primary instrument
table(data_intro$instrument_primary)
data_intro$instrument_primary =
  case_when(
    str_detect(data_intro$instrument_primary, "guitar") ~ "guitar",
    str_detect(data_intro$instrument_primary, "vocal") |
      data_intro$instrument_primary == "singer" ~ "vocals",
    str_detect(data_intro$instrument_primary, "drum") |
      data_intro$instrument_primary == "percussion" ~ "drums",
    str_detect(data_intro$instrument_primary, "bass") ~ "bass",
    data_intro$instrument_primary == "keyboards" ~ "keyboard",
    TRUE ~ data_intro$instrument_primary)
#instrument secondary
table(data_intro$instrument_secondary)
data_intro$instrument_secondary =
  case_when(
    data_intro$instrument_secondary == "keyboards" ~ "keyboard",
    data_intro$instrument_secondary == "" ~ NA,
    TRUE ~ data_intro$instrument_secondary)

#cleaning up the dataset
#moving the new cleaned instrument variables to be after "musician_name"
data_intro = relocate(data_intro, c(instrument_primary, instrument_secondary),
                      .after = musician_name)
#removing old instrument_role variable
data_intro = select(data_intro, -instrument_role)
```

Advanced Code

```r
#separating cells with two instruments into two separate columns
#using str_split_fixed from stringr
data_adv[c("instrument_primary", "instrument_secondary")] =
  str_split_fixed(data_adv$instrument_role, "/", n = 2)

data_adv = data_adv %>%
    #cleaning up instrument labels for using str_detect() from stringr
    mutate(across(c(instrument_primary,
                    instrument_secondary),
                  ~ case_when(
                    str_detect(., "guitar") ~ "guitar",
                    str_detect(., "vocal") | . == "singer" ~ "vocals",
                    str_detect(., "drum") | . == "percussion" ~ "drums",
                    str_detect(., "bass") ~ "bass",
                    . == "keyboards" ~ "keyboard",
                    . == "" ~ NA,
                    T ~ .))) %>%
  #cleaning up the dataset
  #moving the new cleaned instrument variables to be after "musician_name"
  relocate(c(instrument_primary, instrument_secondary),
           .after = musician_name) %>%
  #removing old instrument_role variable
  select(-instrument_role)
```

Your turn!



Mötley Crüe

- **Follow the four-step process**
  - Not required to write code
  - Focus on base R, dplyr, and stringr
- **Pick someone to present your answers**

Part IV: Cleaning the columns

Your turn!

musician_name	nationality
Ritchie Blackmore	England
John Paul Jones	England
David Coverdale	English
James Hetfield	United States
Ronnie James Dio	America
Jimmy Page	United Kingdom
Phil Rudd.	Australia
Steve Harris	
Ian Gillan	England
Robert Plant.	U.k.
Roger Glover.	United Kingdom
David Coverdale	England
Tony Iommi	U.k.
Ian Paice	England
Ronnie James Dio.	English

Group #1
musician_name **and** ***nationality***

age
0
80
74
N/A
67
82
72
24
70
81
78
80
74
78
78
67

Group #2
age

alive	rock_hall_inductee
1	true
y	true
1	no
yes	true
n	y
yes	yes
alive	yes
deceased	yes
yes	yes
1	y
y	true
yes	true
alive	yes
yes	yes
alive	true

Group #3
alive **and** ***rock_hall_inductee***

band
Deep Purple
Led Zepplin
Whitesnake
Metallica
Dio
Led Zeppelin
Ac/Dc
Metallica
Iron Maiden
Black Sabbath
Led Zepplin
Deep Purple
Deep Purple
Black Sabbath

Group #4
band

Part IV: Cleaning the columns

Groups Present

STEP 1: View the data

STEP 2: Verbalize the problems and patterns

STEP 3: Verbalize the fix

STEP 4: Find the code for the fix

For Those About to Rock

(Dawn's Recommendations)



The Basics

- **Led Zeppelin** – *Led Zeppelin III*
- **Deep Purple** – *When We Rock, We Rock, and When We Roll, We Roll*
- **Black Sabbath** – *Heaven and Hell*

Old School

- **Rush**
- **Scorpions** – *Animal Magnetism*
- **Cinderella** – *Night Songs*
- **Mötley Crüe**
- **Dio** – *Holy Diver*
- **Iron Maiden**

New School

- **Greta Van Fleet** – *From the Fires*
- **Queens of the Stone Age** – *…Like Clockwork*
- **Starbenders** – *The Beast Goes On*
- **Venus Grrrls**
- **Ghost** – *IMPERA*
- **Red** – *End of Silence*
- **A Day to Remember** – *What Separates Me From You*
- **Bad Omens** – *The Death of Peace of Mind*